

MZ-UCA Cloud Academy | AWS Solutions Architect Associate (SAA)

Hands-On Lab: Building a Production-Ready Web Server

Objective

Cloud Architects do not just launch servers—they build reliable, secure, and production-ready systems.

In this lab, you will deploy an EC2 instance, configure security, install a web server, assign a static public IP, and validate that your solution is ready for business use.

By completing this lab, you will gain practical experience with:

- ✓ Amazon EC2
 - ✓ Security Groups
 - ✓ Elastic IP Addresses
 - ✓ Linux Administration
 - ✓ Nginx Web Server
 - ✓ Production Readiness Validation
-

Scenario

You have been hired as a Junior Cloud Architect at MZ-UCA Cloud Academy.

The company needs a public-facing website hosted on AWS.

Your responsibility is to deploy, secure, and validate the web server before it can be handed over to the business.

Step 1: Launch an EC2 Instance

Navigate to:

AWS Console → EC2 → Launch Instance

Configure the following:

Name:

mz-uca-web-server

AMI:

Amazon Linux 2023

Instance Type:

t2.micro

Storage:

8 GB

Create or select an existing Key Pair.

Create a new Security Group.

Launch the instance.

Step 2: Configure Security Group Rules

Your web server must allow:

Type	Port
SSH	22
HTTP	80

Type	Port
HTTPS	443

Source:

```
0.0.0.0/0
```

Save the configuration.

Step 3: Connect to Your Server

Using SSH:

```
ssh -i your-key.pem ec2-user@PUBLIC-IP
```

If successful, you should see the Amazon Linux terminal.

Step 4: Update the Operating System

Run:

```
sudo dnf update -y
```

This ensures your server has the latest security patches and updates.

Wait for completion before proceeding.

Step 5: Install Nginx

Install the web server:

```
sudo dnf install nginx -y
```

Start the service:

```
sudo systemctl start nginx
```

Enable automatic startup after reboot:

```
sudo systemctl enable nginx
```

Verify status:

```
sudo systemctl status nginx
```

Expected Result:

```
active (running)
```

Step 6: Test the Default Nginx Website

Open a browser and visit:

```
http://PUBLIC-IP
```

You should see the default Nginx page.

If you can access the page, your server is successfully serving web content.

Step 7: Create Your Own Website

Navigate to:

```
cd /usr/share/nginx/html
```

Remove the default page:

```
sudo rm index.html
```

Create a new page:

```
sudo nano index.html
```

Paste the following content:

```
<!DOCTYPE html>
<html>
<head>
<title>MZ-UCA Cloud Academy</title>
</head>
<body>
<h1>Welcome to MZ-UCA Cloud Academy</h1>
<h2>AWS Solutions Architect Associate Lab</h2>
<p>Production-Ready Web Server Successfully Deployed.</p>
</body>
</html>
```

Save the file.

Step 8: Verify Your Custom Website

Refresh your browser.

Expected Result:

You should now see your custom MZ-UCA Cloud Academy webpage.

Step 9: Allocate an Elastic IP Address

Navigate to:

EC2 → Elastic IPs

Click:

Allocate Elastic IP

Associate the Elastic IP with:

```
mz-uca-web-server
```

Why is this important?

A normal public IP changes when an instance is stopped and started.

An Elastic IP remains fixed, making it suitable for production environments.

Step 10: Validate Auto-Start Functionality

Reboot your server:

```
sudo reboot
```

Reconnect after a few moments.

Verify Nginx is still running:

```
systemctl status nginx
```

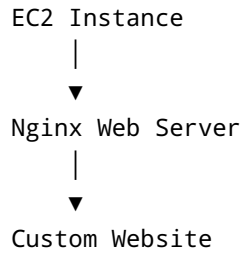
Expected Result:

```
active (running)
```

This confirms your web server automatically starts after a reboot.

Final Architecture

```
graph TD
    Internet --> ElasticIP[Elastic IP]
    ElasticIP --> SecurityGroup[Security Group]
    SecurityGroup --> Instance[ ]
```



Deliverables

Submit the following screenshots:

 EC2 Instance Running

 Security Group Rules

 Nginx Status Showing "Active (Running)"

 Custom Website in Browser

 Elastic IP Successfully Attached

Success Criteria

Before submission, verify:

- ☐ EC2 Running
- ☐ Security Group Configured
- ☐ SSH Access Working
- ☐ Nginx Installed
- ☐ Website Accessible
- ☐ Custom Page Created
- ☐ Elastic IP Attached
- ☐ Auto-Start Validated

Architect's Reflection

After completing the lab, answer the following:

1. Why is an Elastic IP important in production?
2. What role does a Security Group play?
3. Why do we enable Nginx on boot?
4. What would happen if Port 80 was not open?
5. How would you improve this architecture for high availability?

 MZ-UCA Cloud Academy

AWS Solutions Architect Associate (SAA)

Building Future Cloud Architects Through the Unified Control Architecture (UCA)